MANAN RASTOGI

Applied AI Engineer | Computer Vision, Full-Stack Systems, LLM Integration

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EDUCATION

Vellore Institute of Technology (VIT), Vellore, India — **2024 – Present**

B.Tech in Electronics & Communication Engineering (Biomedical Specialization)

IIT Madras, Online, India — **2025 – Present**

BS in Data Science & Applications

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript.
AI/ML & CV: OpenCV, SciPy, NumPy, TrackPy, OCR, LLM APIs (NVIDIA LLaMA 3.1).
Frontend: React, Next.js, Three.js, Tailwind CSS, Framer Motion.
Backend & Databases: Flask, Node.js, REST APIs, JWT, MongoDB, SQLite, SQL.
Infrastructure & Tools: Docker, CI/CD, Vercel, MongoDB Atlas, Git, GitHub, VS Code.

INDEPENDENT DEVELOPMENT

WoundTrack AI: Applied Computer Vision Pipeline — 2024 – Present

Python, OpenCV, Flask, SciPy, SQLite — **Live Demo | GitHub**

- Engineered a CV pipeline tracking **441 cells** across **800+ frames** in **<10 min** on consumer hardware, achieving **28% higher precision** than manual ImageJ workflows.
- Designed LoG blob detection + LAP gap-closing tracker with Kalman-filter prediction, bridging detection gaps to reduce trajectory fragmentation during cell occlusions.
- Built automated evaluation pipeline computing MSD exponents, velocity distributions, and fractal metrics across **5** experimental conditions (**249** control vs. **192** Hg-treated cells).
- Shipped Flask + SQLite dashboard auto-generating per-experiment reports with kinetic curves and one-click trajectory exports, replacing manual spreadsheet analysis.

VITalize AI: Full-Stack Academic Platform — 2025 – Present

Next.js 15, TypeScript, MongoDB, REST APIs, JWT, NVIDIA LLaMA 3.1, Tesseract OCR — **Live Demo**

- Developed a unified academic platform indexing **8,800+** past exam papers with full-text search and subject-level filtering across FFCS planner, paper vault, and exam generator modules.
- Integrated **NVIDIA LLaMA 3.1** with Tesseract OCR fallback pipeline to auto-generate mark-weighted AI solutions from scanned exam PDFs with KaTeX-rendered math.
- Engineered constraint-based **FFCS timetable solver** using pruned backtracking algorithm, resolving clash/faculty/gap constraints across **2,400+** course combinations with sub-second solve time.
- Deployed on **Vercel** with **MongoDB Atlas**; serverless architecture handling concurrent academic workloads.
- Built Chrome extension (Manifest V3) auto-syncing academic data from VIT's VTOP portal.

Interactive 3D Engineering Portfolio — 2024 – Present

TypeScript, React, Three.js (R3F), Framer Motion, Tailwind CSS — **Live Demo | GitHub**

- Architected a React + TypeScript portfolio with lazy-loaded bundle splitting, sub-1s initial load, and responsive accessibility-first layout.
- Implemented WCAG accessibility: keyboard navigation, ARIA labels, skip-to-content, reduced-motion support, and focus management (Lighthouse: 100 SEO, 97 accessibility, 100 best practices).
- Reduced layout shift and animation overhead by implementing Framer Motion with CSS-based reduced-motion fallbacks while maintaining 60 FPS on mobile devices.